Result | Size | Time | Cycles | GPU | GK Frequency | Process | Attributes
Current | 0% - gemm_wmma_kernel | 0 B, 0 B | 14,125 K, 0 | 10.3 ms | 1.13 GB | 0 | NVIDIA GeForce RTX 4060 Laptop GPU | 1.51 GHz | [1662] gemm_wmma_kernel_double

Summary | Details | Source | Disassembly | Command Lists | Raw | Session

Compare | Tools | View | Export

## GPU Speed Of Light Throughput

Roofline Tensor Core

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of the unit. Depending on the limiting factors for the kernel, improving the achieved utilization of the compute and memory pipelines may lead to an increase in performance.

| | |
|---|---|
| Compute (SM) Throughput [%] | 75.24 |
| Memory Throughput [%] | 27.59 |
| L1/TEX Cache Throughput [%] | 77.24 |
| L2 Cache Throughput [%] | 27.59 |
| DRAM Throughput [%] | 8.85 |
| Duration [msec] | 14.13 |
| Elapsed Cycles [cycle] | 13,5968 |
| SM Frequency [Ghz] | 1.13 |
| DRAM Frequency [Ghz] | 7.98 |

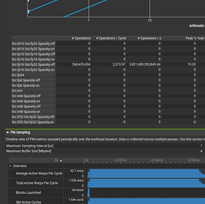
**Small Grid** - The kernel grid is too small to fill the available resources on this device, resulting in only 0.9 full waves across all SMs. Look at Launch Statistics for more details.

Key Performance Indicators

## Roofline Analysis

The ratio of peak float (FP32) to double (FP64) performance on this device is 64:1. The workload achieved close to 0% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the Profiling Guide for more details on roofline analysis.

Tensor Core Operations Roofline




| | # Operations | # Operations / Cycle | # Operations / s | Peak % Val Operations / Cycle | Peak Operations / s |
|---|---|---|---|---|---|
| fp16 | 0 | 0 | 0 | 0.00 | 0 |
| fp16 fp32 fp32 Sparsity off | 0 | 0 | 0 | 0.00 | 0 |
| fp16 fp32 fp32 Sparsity on | 0 | 0 | 0 | 0.00 | 0 |
| fp16 fp32 fp32 Sparsity off | 0 | 0 | 0 | 0.00 | 0 |
| fp16 fp32 fp32 Sparsity on | 268,435,456 | 2,019.97 | 3,821,499,290,648.68 | 19.33 | 19,780,189,693,951.14 |
| bf16 fp32 fp32 Sparsity off | 0 | 0 | 0 | 0.00 | 0 |
| bf16 fp32 fp32 Sparsity on | 0 | 0 | 0 | 0.00 | 0 |
| tf32 fp32 fp32 Sparsity off | 0 | 0 | 0 | 0.00 | 0 |
| tf32 fp32 fp32 Sparsity on | 0 | 0 | 0 | 0.00 | 0 |
| fp64 | 0 | 0 | 0 | 0.00 | 0 |
| fp64 fp64 fp64 Sparsity off | 0 | 0 | 0 | 0.00 | 0 |
| fp64 fp64 fp64 Sparsity on | 0 | 0 | 0 | 0.00 | 0 |
| int8 int8 int32 Sparsity off | 0 | 0 | 0 | 0.00 | 0 |
| int8 int8 int32 Sparsity on | 0 | 0 | 0 | 0.00 | 0 |
| tcu tf32 fp32 fp32 Sparsity off | 0 | 0 | 0 | 0.00 | 0 |
| tcu tf32 fp32 fp32 Sparsity on | 0 | 0 | 0 | 0.00 | 0 |

## GPU Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over time.

| | |
|---|---|
| Maximum Sampling Interval [us] | 4 Pass Groups |
| Minimum Sampling Interval [us] | 2 |

## Compute Workload Analysis

Pipe Utilization (Elapsed Cycles)

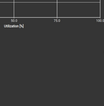
Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

| | |
|---|---|
| Executed Ipc Elapsed [inst/cycle] | 07.83 |
| Issued Ipc Active [inst/cycle] | 07.87 |
| Issued Ipc Elapsed [inst/cycle] | 07.83 |
| SM Busy [%] | 19 |

Key Performance Indicators

Pipe Utilization (% of elapsed cycles)

Pipe Utilization (% of peak instructions executed over elapsed cycles)




## Memory Workload Analysis

Memory Chart

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed charts of the memory units. Detailed tables with data for each memory unit.

| | | | |
|---|---|---|---|
| Memory Throughput [Gbyte/s] | 67.44 | Mem Busy [%] | 25.24 |
| L1/TEX Hit Rate [%] | 34.00 | Max Bandwidth [%] | 27.59 |
| L2 Hit Rate [%] | 79.40 | Mem Pipes Busy [%] | 27.59 |
| L2 Compression Input Sectors [sector] | 0 | Local Memory Spilling Requests | 0 |
| L2 Compression Ratio | 0 | Local Memory Spilling Request Overhead [%] | 0 |
| L2 Compression Success Rate [%] | 0 | L2 Promoting Size [Byte] | 8,58 |

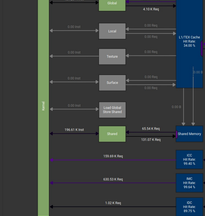
**L1TEX Global Load Access Pattern** - The memory access pattern for global loads from L1TEX might not be optimal. On average, only 20.2 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the Source Counters section for uncoalesced global loads.
Est. Speedup: 37.81%

**L1TEX Global Store Access Pattern** - The memory access pattern for global stores to L1TEX might not be optimal. On average, only 16.0 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the Source Counters section for uncoalesced global stores.
Est. Speedup: 12.72%

Key Performance Indicators

Memory Chart

Values: Transfer Size | Inactivity | Grayed Out


Global
Local
Texture
Surface
Load Global Store Shared
Shared
Device Memory
L1/TEX Cache
L2 Cache
Compression
SM
Memory
System Memory
135.26 K Inst
135.17 K Req
4.15 K Req
4.15 K Req
4.15